LW3: Merged Data, DAX, and DAX Function Insights

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IS 107 - Business Intelligence

Part 1: Review the Merged Dataset

OrderID	CustomerID	OrderDate	SalesAmount	CustomerName	City	Region
O003	C137	Tuesday, March 11, 2025	800	Customer 3	Butuan	Mindanao
O007	C091	Friday, February 7, 2025	1200	Customer 7	Zamboanga	Mindanao
O009	C027	Saturday, March 29, 2025	45000	Customer 9	Cagayan de Oro	Mindanao
O010	C094	Wednesday, February 19, 2025	1200	Customer 10	Cagayan de Oro	Mindanao
O012	C146	Saturday, March 1, 2025	3000	Customer 12	Cagayan de Oro	Mindanao
O013	C131	Friday, March 7, 2025	1200	Customer 13	Cagayan de Oro	Mindanao
O016	C173	Sunday, March 2, 2025	45000	Customer 16	Butuan	Mindanao
O024	C030	Monday, February 17, 2025	1200	Customer 24	Cagayan de Oro	Mindanao
O025	C100	Friday, March 28, 2025	2200	Customer 25	Davao	Mindanao
O028	C127	Friday, January 10, 2025	800	Customer 28	Cagayan de Oro	Mindanao
O030	C122	Tuesday, March 11, 2025	3000	Customer 30	Davao	Mindanao
O036	C077	Saturday, March 29, 2025	3000	Customer 36	Davao	Mindanao
O042	C132	Wednesday, March 19, 2025	3000	Customer 42	Zamboanga	Mindanao
O043	C020	Wednesday, February 5, 2025	3000	Customer 43	Cagayan de Oro	Mindanao
O044	C024	Tuesday, March 4, 2025	800	Customer 44	Zamboanga	Mindanao
O047	C046	Wednesday, January 22, 2025	800	Customer 47	Zamboanga	Mindanao
O049	C023	Wednesday, January 8, 2025	1200	Customer 49	Zamboanga	Mindanao
O054	C018	Tuesday, January 7, 2025	1500	Customer 54	Cagayan de Oro	Mindanao
O058	C162	Monday, March 24, 2025	7500	Customer 58	Butuan	Mindanao
O060	C174	Monday, March 10, 2025	45000	Customer 60	Butuan	Mindanao
O061	C184	Wednesday, March 19, 2025	2200	Customer 61	Cagayan de Oro	Mindanao
O062	C076	Monday, March 24, 2025	2200	Customer 62	Cagayan de Oro	Mindanao
O067	C087	Friday, March 28, 2025	7500	Customer 67	Davao	Mindanao
O075	C079	Wednesday, January 1, 2025	3000	Customer 75	Davao	Mindanao

Table: the Merged_Sales_Data (200 rows)

Data

Search

Customers

the Merged_Sales_Data

Guide Questions

1. Is the dataset in flat-table format?
   - Yes, the dataset is in flat-table format all data is stored in a single table with rows and columns, where each row represents one order.
2. Which columns describe customers?
   - CustomerID, CustomerName, City, and Region describe the customers.
3. Which columns represent transactions?
   - OrderID, OrderDate, and SalesAmount represent the transactions.

Part 2: Identify Fact and Dimension Tables

Create DimCustomer

Queries [4]

- the Merged_Sales_Data
- Customers
- DimCustomer
- FactSales

```
= Table.Distinct(#"Removed Columns", {"CustomerID"})
```

	CustomerID	CustomerName	City	Region
1	C009	Customer 1	Legazpi	Luzon
2	C156	Customer 2	Manila	Luzon
3	C137	Customer 3	Butuan	Mindanao
4	C127	Customer 4	Dumaguete	Visayas
5	C105	Customer 5	Dumaguete	Visayas
6	C054	Customer 6	Cebu	Visayas
7	C091	Customer 7	Zamboanga	Mindanao
8	C016	Customer 8	Baguio	Luzon
9	C027	Customer 9	Cagayan de Oro	Mindanao
10	C094	Customer 10	Cagayan de Oro	Mindanao
11	C146	Customer 12	Cagayan de Oro	Mindanao
12	C131	Customer 13	Cagayan de Oro	Mindanao
13	C170	Customer 14	Baguio	Luzon
14	C021	Customer 15	Tacloban	Visayas
15	C173	Customer 16	Butuan	Mindanao
16	C115	Customer 17	Baguio	Luzon
17	C192	Customer 18	Iloilo	Visayas
18	C159	Customer 19	Baguio	Luzon
19	C026	Customer 21	Manila	Luzon
20	C086	Customer 22	Legazpi	Luzon
21	C051	Customer 23	Tacloban	Visayas
22	C030	Customer 24	Cagayan de Oro	Mindanao
23	C100	Customer 25	Davao	Mindanao
24	C088	Customer 26	Cebu	Visayas

4 COLUMNS, 127 ROWS Column profiling based on top 1000 rows

Create FactSales

Queries [4]

- the Merged_Sales_Data
- Customers
- DimCustomer
- FactSales

```
= Table.RemoveColumns(Source,{"CustomerName", "City", "Region"})
```

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	3000
2	O002	C156	2/12/2025	7500
3	O003	C137	3/11/2025	800
4	O004	C127	1/1/2025	45000
5	O005	C105	3/3/2025	1200
6	O006	C054	3/29/2025	2200
7	O007	C091	2/7/2025	1200
8	O008	C016	1/17/2025	2200
9	O009	C027	3/29/2025	45000
10	O010	C094	2/19/2025	1200
11	O011	C094	2/22/2025	3000
12	O012	C146	3/1/2025	3000
13	O013	C131	3/7/2025	1200
14	O014	C170	2/12/2025	45000
15	O015	C021	1/6/2025	800
16	O016	C173	3/2/2025	45000
17	O017	C115	2/15/2025	45000
18	O018	C192	2/10/2025	800
19	O019	C159	2/5/2025	7500
20	O020	C156	1/28/2025	800
21	O021	C026	1/22/2025	800
22	O022	C086	1/14/2025	800
23	O023	C051	1/14/2025	3000
24	O024	C030	2/17/2025	1200

4 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

Guide Questions

1. Why must dimension tables contain unique keys?

   - Dimension tables must have unique keys so that each record can be correctly identified and matched when linking to the fact table. Without unique keys, relationships between tables will be inaccurate or won't work at all.

2. What problems occur if duplicates exist in DimCustomer?

   - If duplicates exist in DimCustomer, Power BI won't be able to create a proper one-to-many relationship with FactSales, causing errors in reports and incorrect results for example, a customer's sales being counted multiple times.

Part 3: Define Primary & Foreign Keys

Identify Keys

Table	Key	Type
DimCustomer	CustomerID	Primary Key
FactSales	CustomerID	Foreign Key
FactSales	OrderID	Primary Key

Guide Questions

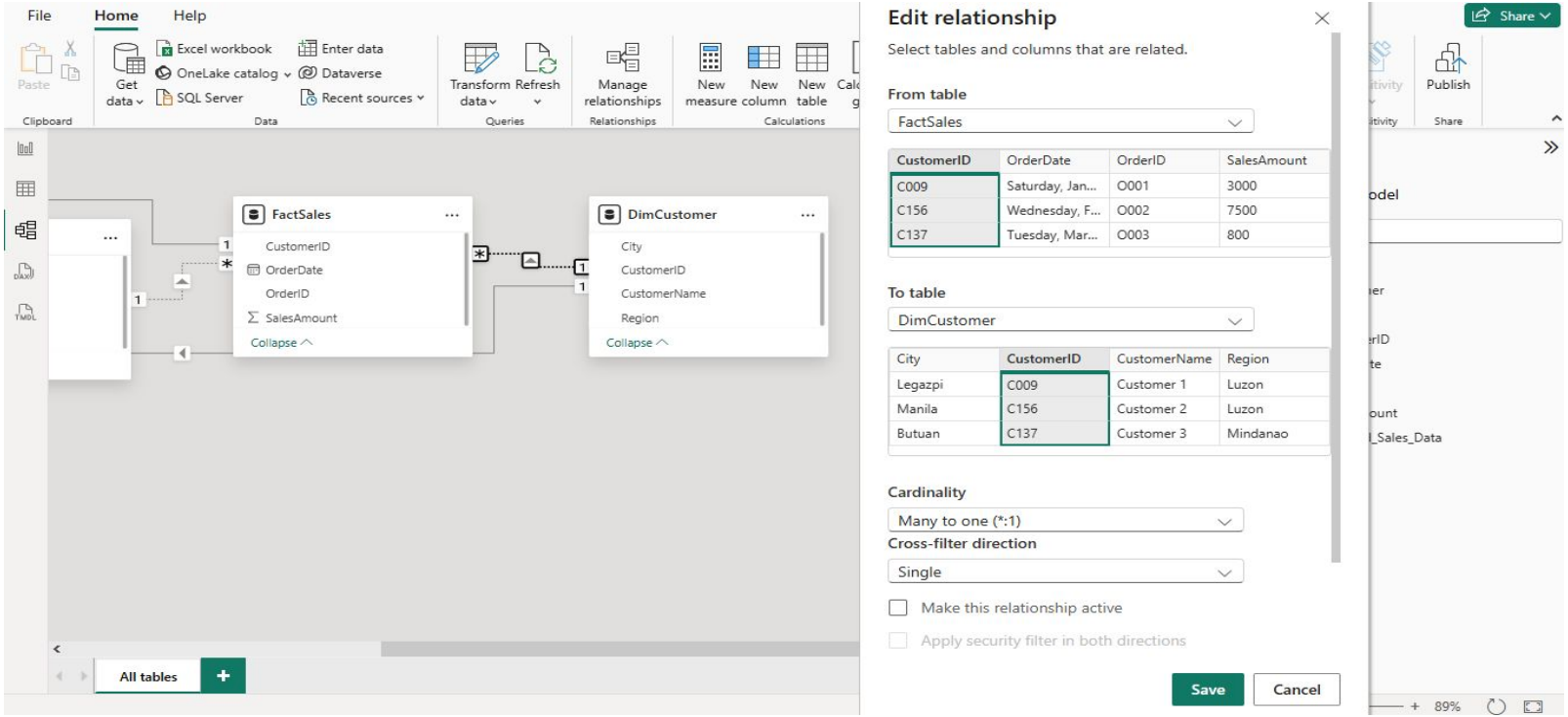
1. What is a primary key?

   - A primary key is a column that uniquely identifies each row in a table. In the image, CustomerID is the primary key in DimCustomer and OrderID is the primary key in FactSales meaning no two rows can have the same value in those columns.

2. Why is CustomerID a foreign key in FactSales?

   - CustomerID is a foreign key in FactSales because it doesn't uniquely identify rows in that table instead, it references the primary key in DimCustomer to link each sales transaction to its corresponding customer. It creates the relationship between the two tables.

Part 4: Create Relationships



Guide Questions

1. Why should relationships flow from dimension to fact?
   - So that filters from dimension tables (like customer or region) correctly flow into the fact table, allowing accurate slicing and analysis of sales data.
2. What happens if the relationship is Many-to-Many?
   - It causes ambiguous matches, leading to incorrect totals, double counting, and unreliable reports.

Part 5: Validate the Data Model

CustomerName	Region	Sum of SalesAmount
Customer 1	Luzon	15700
Customer 10	Mindanao	5400
Customer 100	Mindanao	48000
Customer 101	Mindanao	3000
Customer 102	Luzon	800
Customer 106	Visayas	1500
Customer 108	Mindanao	1200
Customer 110	Visayas	45000
Customer 111	Mindanao	4200
Customer 113	Luzon	8700
Customer 114	Mindanao	46500
Customer 115	Mindanao	45000
Customer 12	Mindanao	3000
Customer 120	Visayas	2200
Customer 123	Mindanao	7500
Customer 124	Visayas	45000
Customer 128	Mindanao	1200
Customer 129	Visayas	3000
Customer 13	Mindanao	53700
Customer 134	Visayas	45000
Customer 137	Mindanao	10900
Customer 139	Luzon	45000
Customer 14	Luzon	45000
Customer 140	Visayas	3000
Customer 141	Visayas	4500
Customer 142	Mindanao	45800
Customer 145	Visayas	7500
Customer 146	Visayas	3000
Total		**2213800**

Page 1

Filters

Search

Filters on this visual

CustomerName
is (All)

Region
is (All)

Sum of SalesAmou...
is (All)

Add data fields here

Filters on this page

Add data fields here

Filters on all pages

Visualizations

Build visual

Columns

CustomerName

Region

Sum of SalesAmount

Drill through

Cross-report Off

Keep all filters On

Data

Search

Customers

DimCustomer

City

CustomerID

CustomerName

Region

FactSales

CustomerID

OrderDate

OrderID

SalesAmount

the Merged_Sales_Data

Page 1 of 1

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Guide Questions

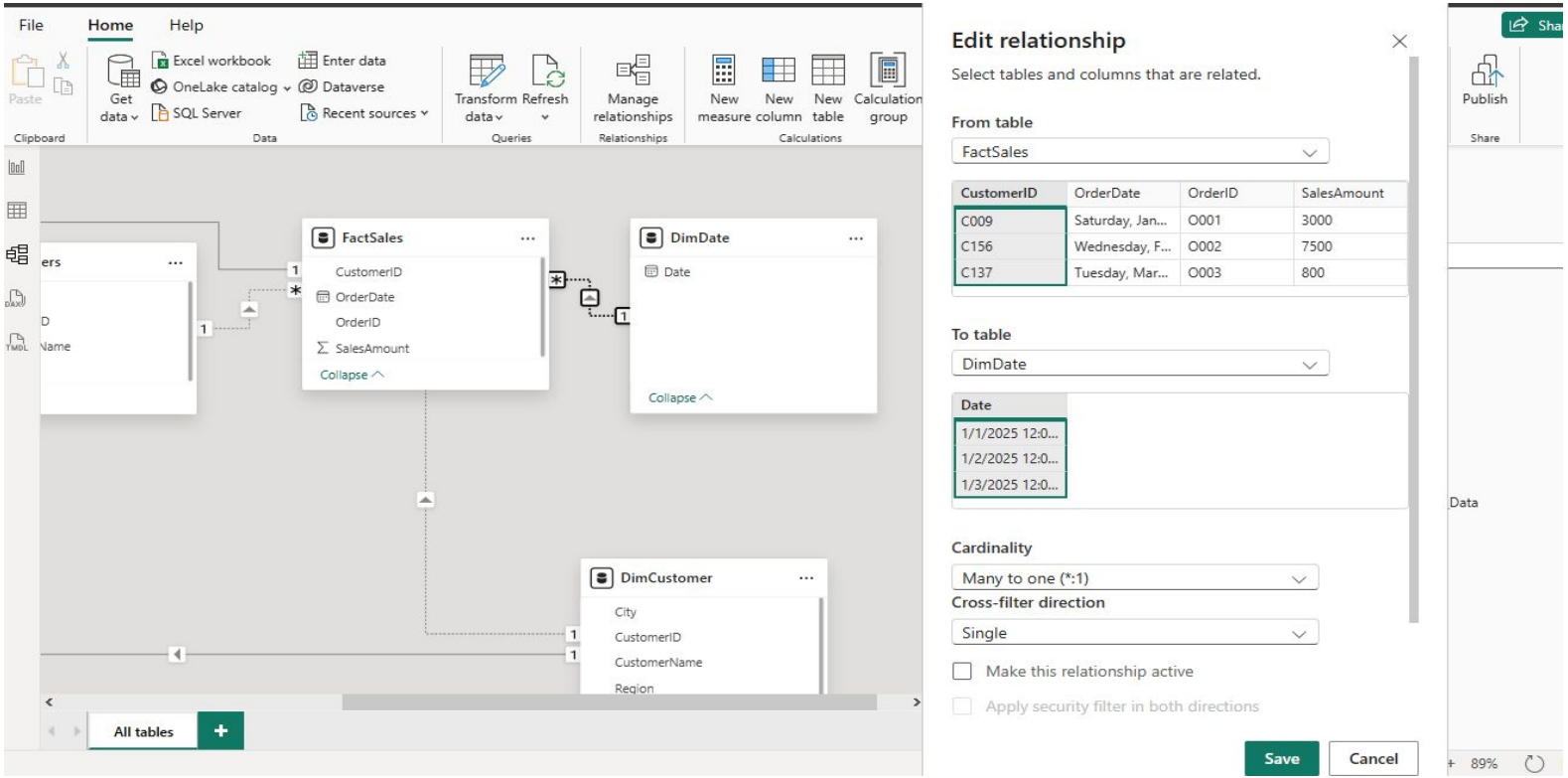
1. Did sales aggregate correctly per customer?

- Yes, sales aggregated correctly each customer shows their total SalesAmount, meaning the relationship between DimCustomer and FactSales is working properly.

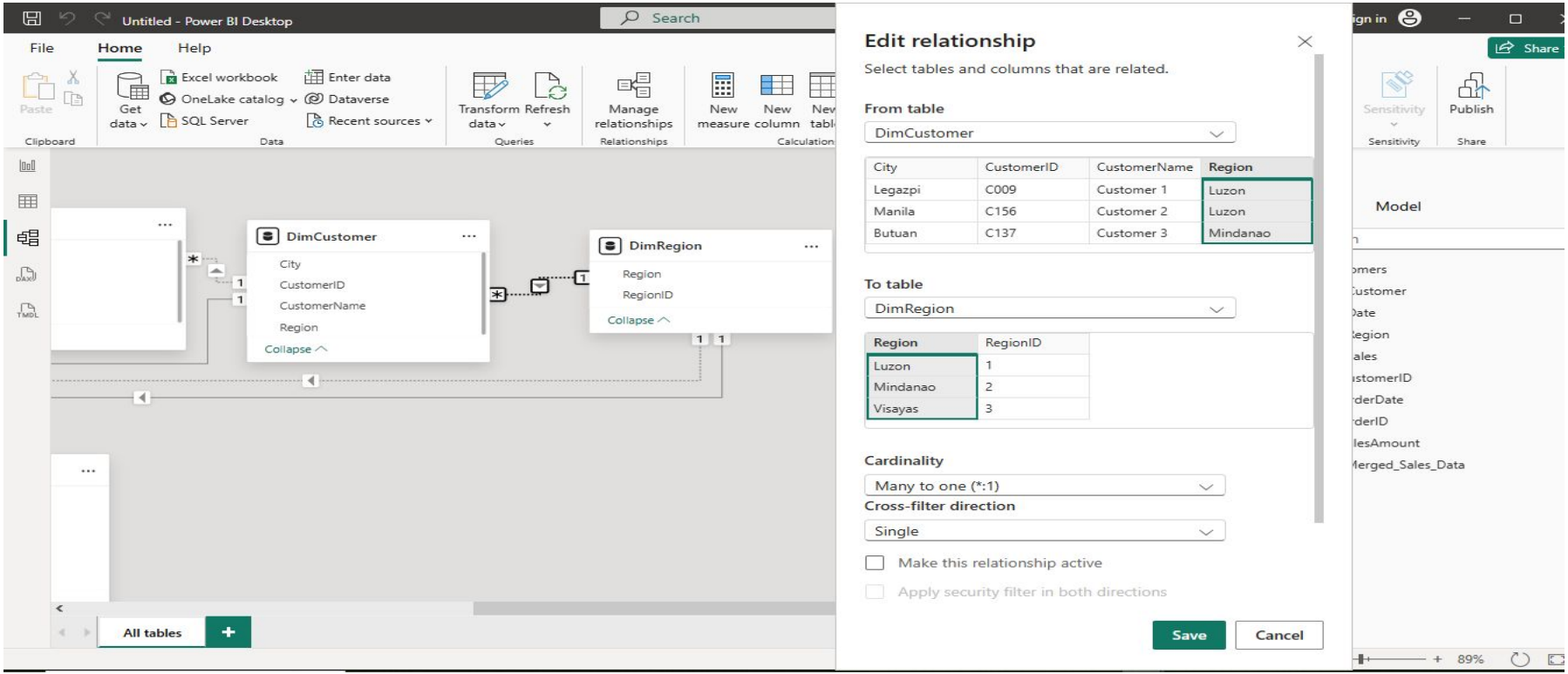
2. What would happen if relationships were missing?

- Without relationships, SalesAmount would show the same grand total for every customer instead of their individual totals, making the report completely inaccurate.

Enhancement Activities

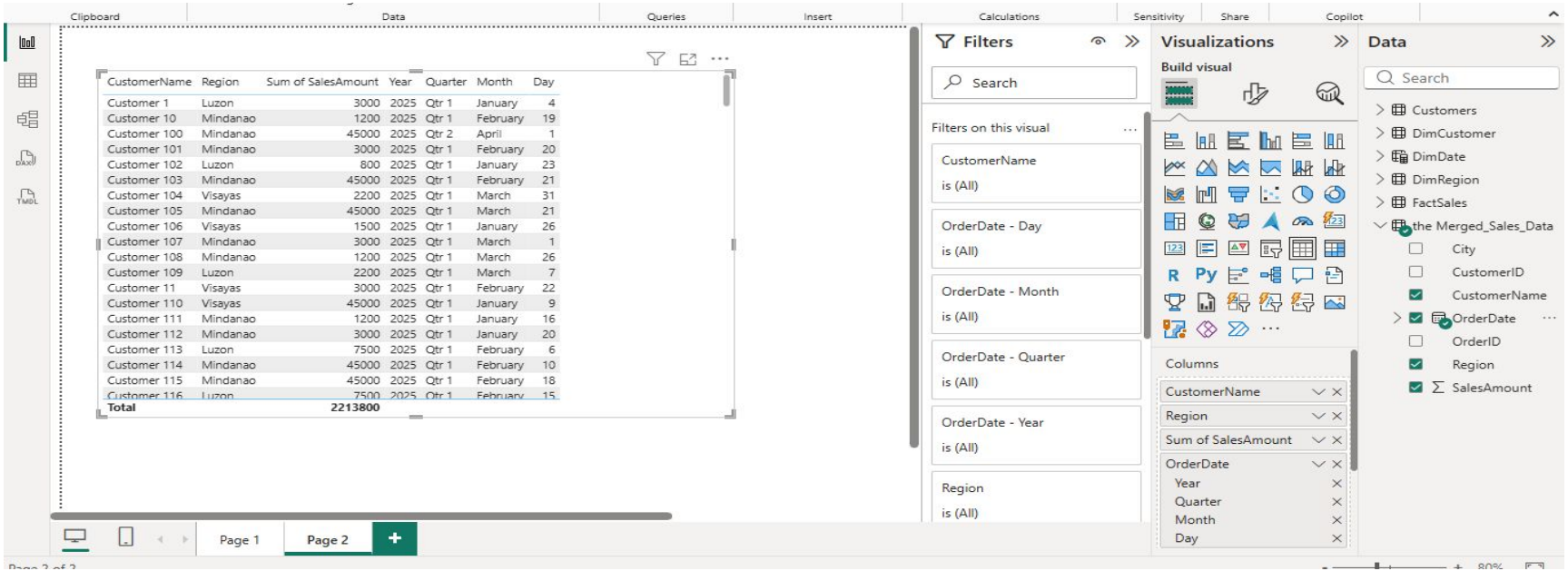


Enhancement 2: Create a DimRegion Table



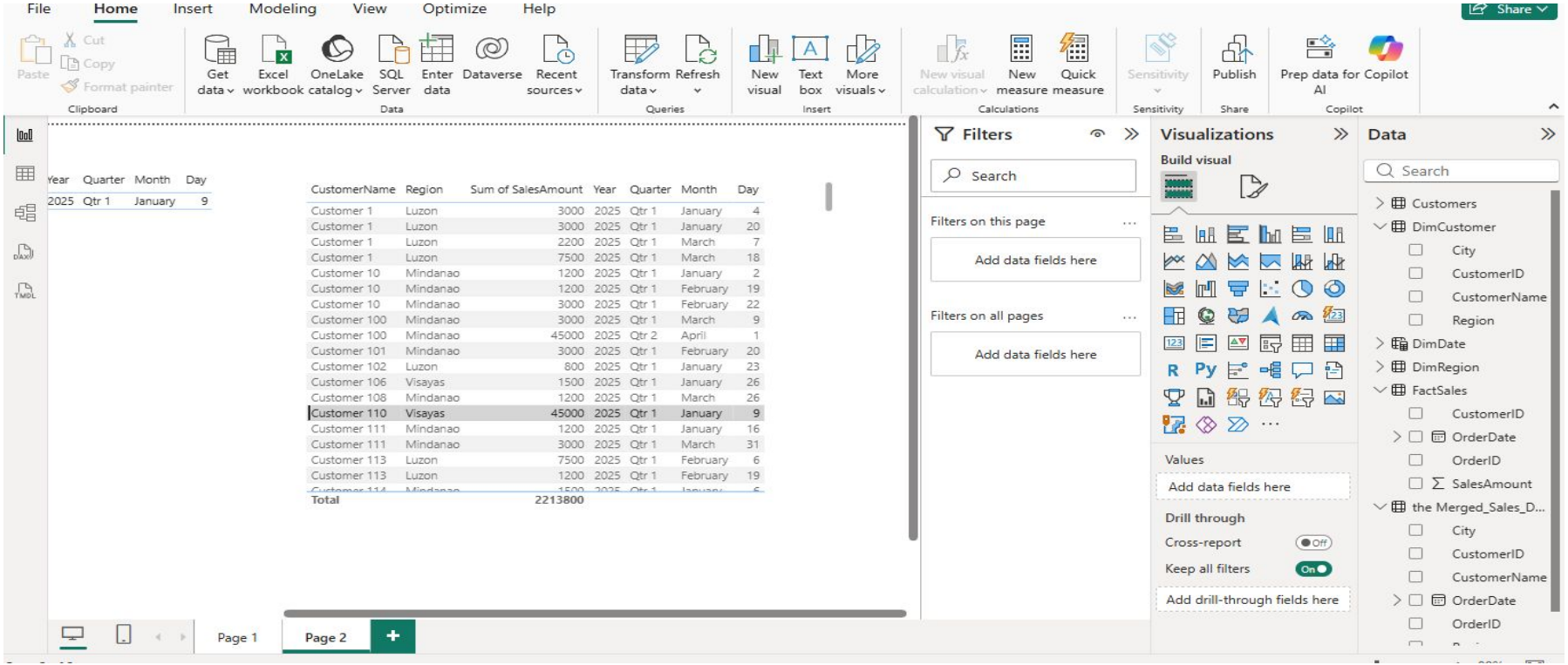
Enhancement 3: Compare Flat vs Star Schema Performance

- Create visuals using a flat table



CustomerName	Region	Sum of SalesAmount	Year	Quarter	Month	Day
Customer 1	Luzon	3000	2025	Qtr 1	January	4
Customer 10	Mindanao	1200	2025	Qtr 1	February	19
Customer 100	Mindanao	45000	2025	Qtr 2	April	1
Customer 101	Mindanao	3000	2025	Qtr 1	February	20
Customer 102	Luzon	800	2025	Qtr 1	January	23
Customer 103	Mindanao	45000	2025	Qtr 1	February	21
Customer 104	Visayas	2200	2025	Qtr 1	March	31
Customer 105	Mindanao	45000	2025	Qtr 1	March	21
Customer 106	Visayas	1500	2025	Qtr 1	January	26
Customer 107	Mindanao	3000	2025	Qtr 1	March	1
Customer 108	Mindanao	1200	2025	Qtr 1	March	26
Customer 109	Luzon	2200	2025	Qtr 1	March	7
Customer 11	Visayas	3000	2025	Qtr 1	February	22
Customer 110	Visayas	45000	2025	Qtr 1	January	9
Customer 111	Mindanao	1200	2025	Qtr 1	January	16
Customer 112	Mindanao	3000	2025	Qtr 1	January	20
Customer 113	Luzon	7500	2025	Qtr 1	February	6
Customer 114	Mindanao	45000	2025	Qtr 1	February	10
Customer 115	Mindanao	45000	2025	Qtr 1	February	18
Customer 116	Luzon	7500	2025	Qtr 1	February	15
Total		**2213800**				

- Create visuals using a star schema

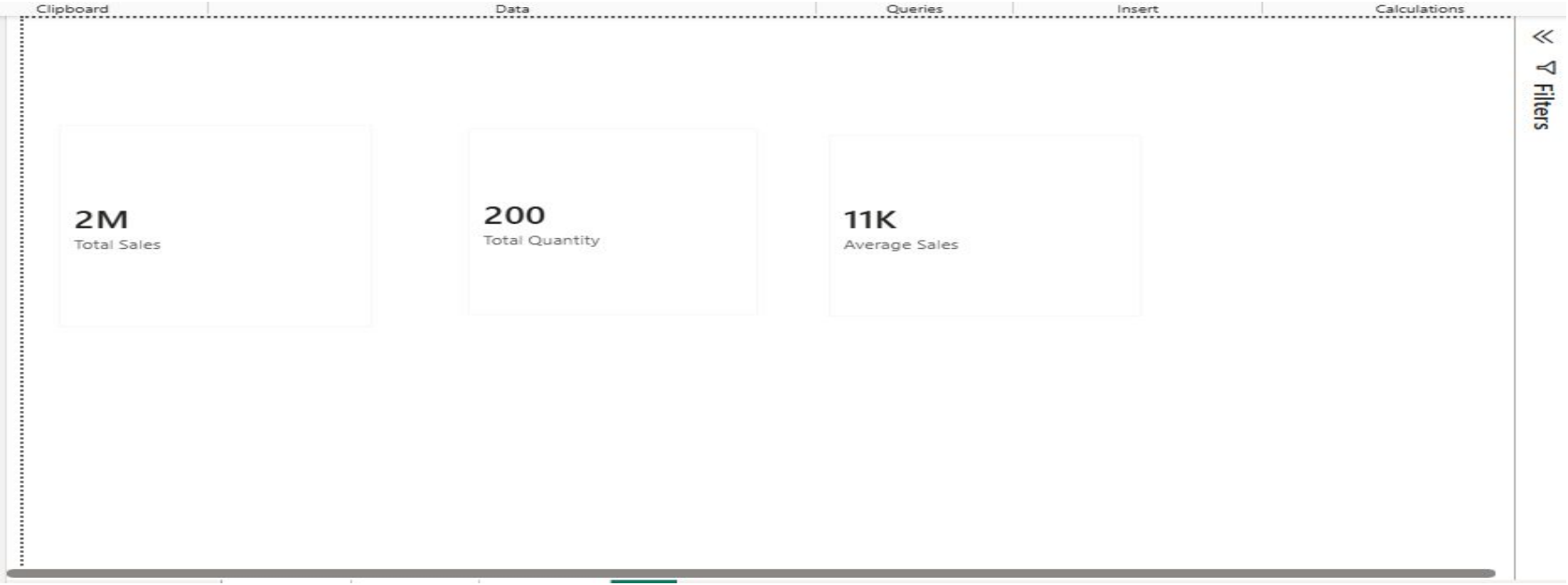


- Observe the performance difference

The star schema visual filters faster and more accurately because data is organized into separate dimension and fact tables, reducing redundancy compared to the flat table.

Introduction to DAX

Create Measures



Guide Questions

1. What is a DAX measure?

   - A DAX measure is a formula written in Data Analysis Expressions (DAX) that calculates values dynamically based on the context of your report, like filters or slicers.

2. Difference between SUM and AVERAGE?

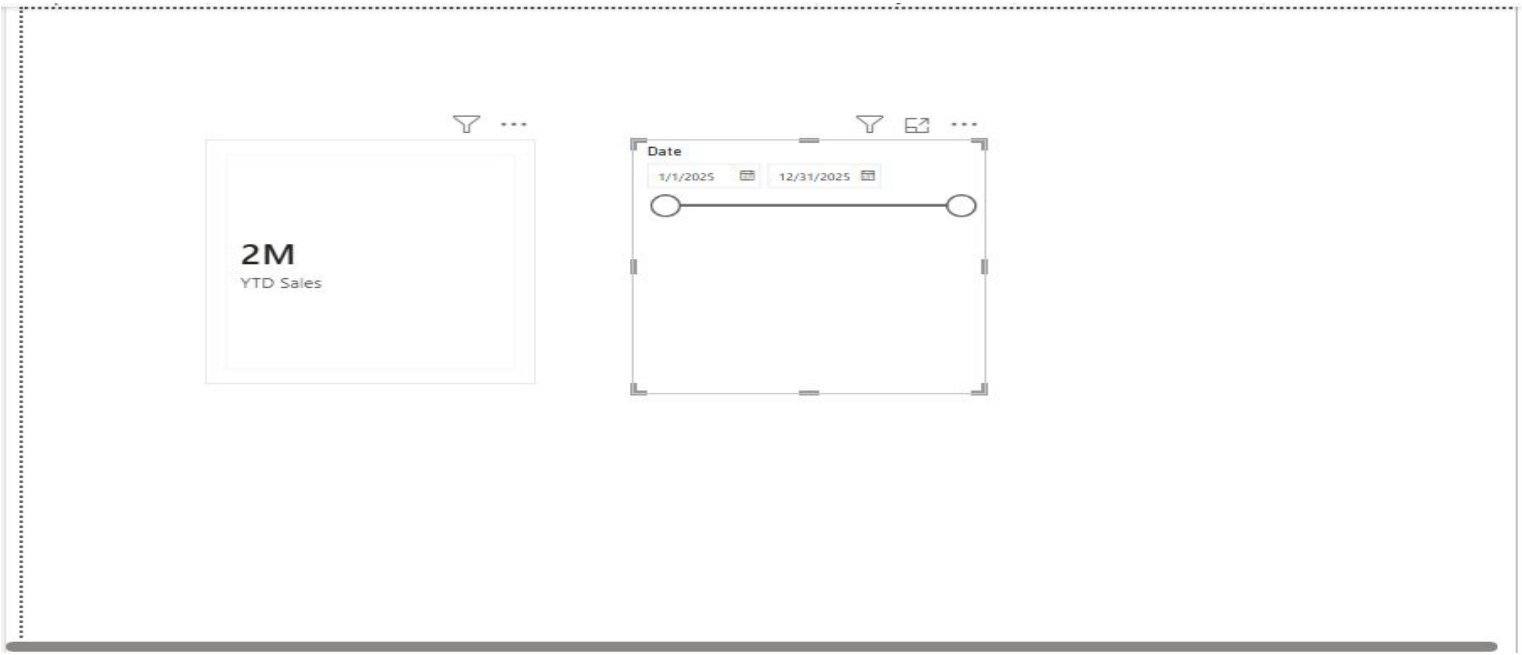
   - SUM adds all values together to give a total, while AVERAGE divides the total by the number of records to give the mean value.

3. Why use measures instead of calculated columns?

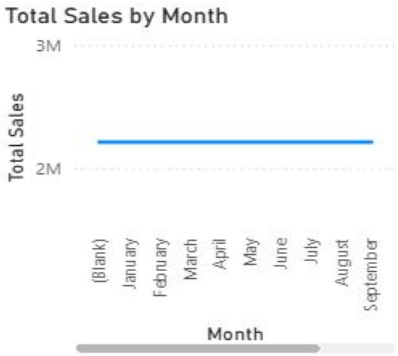
   - Measures calculate results dynamically at report runtime based on filters and context, making them more efficient and flexible. Calculated columns are computed row by row and stored in the table, consuming more memory.

Enhancement Activities DAX Functions

1. Year-to-Date Sales



2. Monthly Sales Growth



Guide Questions

1. What is time intelligence in DAX?

   - Time intelligence in DAX refers to functions that perform calculations based on time periods, such as year-to-date, month-over-month, and previous month comparisons.

2. Why is a date table required?

   - A date table is required because time intelligence functions like TOTALYTD and PREVIOUSMONTH need a continuous, unbroken sequence of dates to calculate correctly across time periods.

3. How does PREVIOUSMONTH help analyze trends?

   - PREVIOUSMONTH retrieves the total from the previous month so you can compare it to the current month, making it easy to see whether sales are growing or declining over time.

4. What insights can YTD Sales provide?

   - YTD Sales shows the cumulative total from the start of the year up to the current date, helping businesses track overall progress toward annual sales targets.